Today

- Born's hypothesis and the Copenhagen interpretation
- Conservation of probability and probability current
- The Schrödinger equation as an operator equation

1.7 Answer to question 3 part I: Statistical interpretation and Born's hypothesis

According to the principles of quantum mechanics, we cannot predict with certainty where the electron falls on the screen. We can only talk about the probability that the electron falls at a given point on the screen. The probability density that the electron falls at point $\vec{r}$ at time t on the screen is given by

$$\mathcal{P}(\vec{r}, t) = |\psi(\vec{r}, t)|^2 = \psi(\vec{r}, t)^* \psi(\vec{r}, t), \tag{1.7.1}$$

where $\psi(\vec{r}, t)$ is the wavefunction. With this, the electron density pattern seen in the figure above becomes the probability density that a single electron falls at that point, where $\psi(\vec{r}, t)$ is the wavefunction for a single electron. The wavefunction is also often called the probability amplitude or the state (or the state vector).

We summarize Born's interpretation and requirement on the wavefunction as follows.

1. Statistical interpretation: The wavefunction is the probability amplitude and $|\psi(\vec{r}, t)|^2$ is the probability density for finding the particle at point $\vec{r}$ at time t.
2. Normalization: Because the particle must be found at some point at a given time, the total probability for finding the particle over all space must be one,

$$\int_{\text{over all space}} |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1. \tag{1.7.2}$$

   The wavefunction must be square-integrable.

   In particular, it must have finite values everywhere, so that the $\int |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1$. If a wavefunction $\psi_s(\vec{r}, t)$ is obtained as a solution of the Schrödinger equation, it is finite everywhere, and $\int |\psi_s(\vec{r}, t)|^2 d^3\vec{r} = A \neq 1$, then we normalize (or rescale) the wavefunction and promote it to $\psi(\vec{r}, t) = \frac{\psi_s(\vec{r}, t)}{\sqrt{A}}$ so that $\int |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1$ and the wavefunction is normalized. If we find solutions to the Schrödinger equation that are not normaliz-

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able, we discard them. However, we will find later in topic 3 what are called improper wavefunctions that are not normalizable in the standard sense defined above, but that are useful.

Question 1.7.1: What is the dimension (or unit) of the wavefunction in three dimensions? What about in one dimension?

Answer: It has unit of $m^{-3/2}$, since $\int |\psi(\vec{r},t)|^2 d^3\vec{r} = 1$ is just a dimensionless number. In one dimension, it has unit of $m^{-1/2}$, since $\int |\psi(x,t)|^2 dx = 1$.

1.8 Answer to question 3 part II: The wavefunction before and after measurement and the Copenhagen interpretation

According to the Copenhagen interpretation (mainly due to Bohr):

1. When the electron is detected, at point A on the screen, the wavefunction collapses from $\psi_{\text{before measurement}}$ to $\psi_{\text{after measurement}}$ that is narrowly peaked around its location as schematically shown in the figure below.

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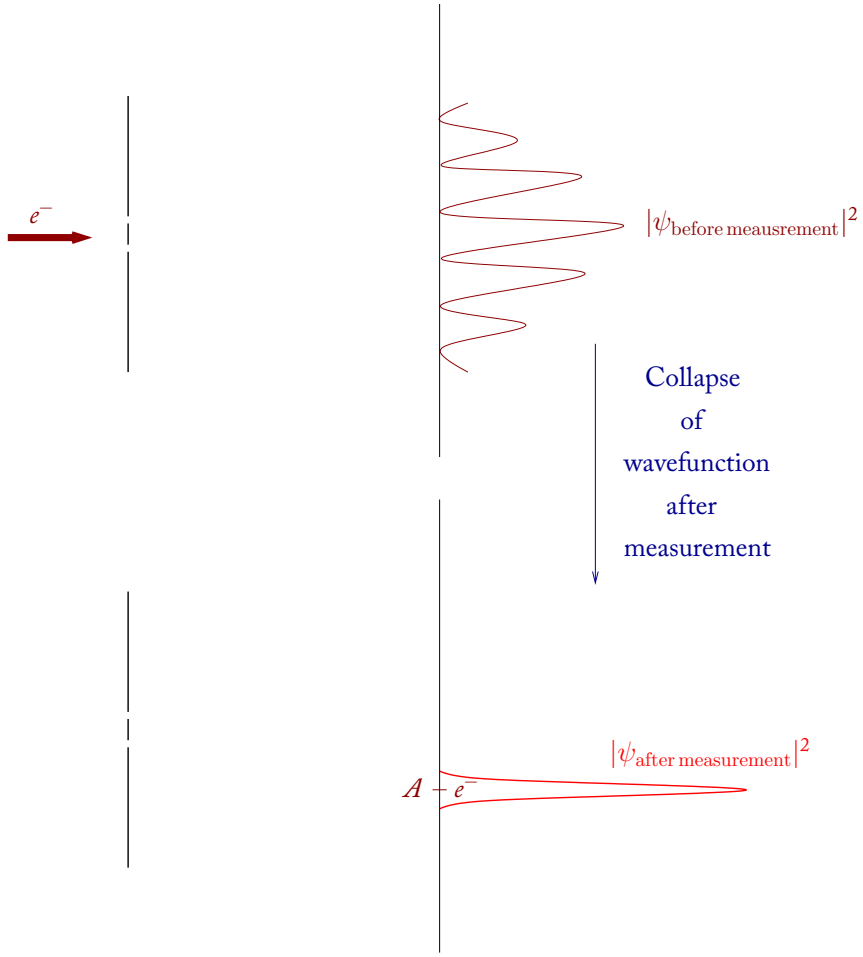


Figure 1.8.1: Collapse of wavefunction after measurement.

We have represented the wavefunction after collapse as a narrowly peaked function that has nonzero values around the point A, because a measuring device could only locate the position within some width. In other words, the act of measurement discontinuously collapses and changes the wavefunction of the electron.

2. The electron does not have a single position before measurement. The electron is represented by the wavefunction. We could say the electron is everywhere on the screen with the wavefunction describing the probability amplitude.

Question 1.8.1: Suppose we made a measurement of the location of the electron fired through the slits and we found it at point A as shown in the figure above. Was the electron at point A before we measured it.

Answer: No. The electron before the measurement is represented by the wave-

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function before collapse, and it is everywhere the wavefunction is nonzero.

The interpretations of quantum mechanics have been a fertile ground for extensive debate among physicists and philosophers. Some of the questions raised by the opponents of the Copenhagen interpretation are the following: probabilistic interpretation, discontinuous collapse of wavefunction, role of observer, and role of measurement device. Various alternative interpretations have been proposed such as hidden variables and many-worlds.

Here we only point out the hidden variables alternative interpretation. According to it, the quantum mechanics in the Copenhagen interpretation is incomplete and there must be hidden variables that need to be included, and the hidden variables carry detailed information, such as about where the electron would land in the diffraction experiment.

It turns out Bell's inequalities (which started with Bell in 1964) provide a setting which allow to make experimental tests between the Copenhagen and the hidden world interpretations, and these experiments support the Copenhagen interpretation and contradict the hidden world interpretation.

Note that the electron before measurement exhibits wave property while its particle property is observed after measurement. It demonstrates only one of its wave or particle property at a single instant. This is an example of what is called the wave-particle complementarity principle proposed by Bohr in 1927. According to the principle, an object can behave either only as a particle or only as a wave, but not both simultaneously, in a single experiment at the same time.

As far as we know, the Schrödinger equation applies to both the electron and the measuring device. We will ignore the interaction of the system such as the electron and the measuring device and focus only on the system and follow Born's and the Copenhagen interpretations.

Question 1.8.2: What do you think happens if we have a detecting devices at the location of the slits that tells us which slit an electron passes through?

Answer:

1.9 Conservation of probability and probability current

If we start with an initial wavefunction that is normalized and evolve it in time using the Schrödinger equation, it continues to stay normalized. In other words, the Schrödinger equation preserves the normalizablity of the wavefunction. We say the Schrödinger equation pro-

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duces unitary time evolution. This is proved below.

The Schrödinger equation for the wavefunction of a particle of mass m and potential energy $V(\vec{r},t)$ in one-dimensional space is

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t). \tag{1.9.1}$$

Taking the complex conjugate of the Schrödinger equation, noting that $V(\vec{r},t)$ is real,

$$-i\hbar\frac{\partial\psi(\vec{r},t)^*}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)^* + V(\vec{r},t)\psi(\vec{r},t)^*, \tag{1.9.2}$$

or $i\hbar\frac{\partial\psi(\vec{r},t)^*}{\partial t} = \frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)^* - V(\vec{r},t)\psi(\vec{r},t)^*$. Using this,

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\mathcal{P} &= i\hbar\frac{\partial}{\partial t}|\psi(\vec{r},t)|^2 = i\hbar\frac{\partial}{\partial t}\left(\psi(\vec{r},t)^*\psi(\vec{r},t)\right) = \psi(\vec{r},t)^*\left(i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t)\right) + \psi(\vec{r},t)\left(i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t)^*\right)\\
&= \psi(\vec{r},t)^*\left(-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(x,t)\psi(\vec{r},t)\right) + \psi(\vec{r},t)\left(\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)^* - V(x,t)\psi(\vec{r},t)^*\right)\\
&= -\frac{\hbar^2}{2m}\left(\psi(\vec{r},t)^*\vec{\nabla}^2\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}^2\psi(\vec{r},t)^*\right)\\
&= -\frac{\hbar^2}{2m}\vec{\nabla}\cdot\left(\psi(\vec{r},t)^*\vec{\nabla}\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}\psi(\vec{r},t)^*\right).
\end{aligned} \tag{1.9.3}$$

Thus,

$$\frac{\partial}{\partial t}\mathcal{P} = \vec{\nabla}\cdot\left[\frac{i\hbar}{2m}\left(\psi(\vec{r},t)^*\vec{\nabla}\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}\psi(\vec{r},t)^*\right)\right] = -\vec{\nabla}\cdot\vec{J}, \tag{1.9.4}$$

where

$$\vec{J} = -\frac{i\hbar}{2m}\left(\psi(\vec{r},t)^*\vec{\nabla}\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}\psi(\vec{r},t)^*\right) \tag{1.9.5}$$

is called probability current. In other words, the probability density $\mathcal{P}$ satisfies the continuity equation,

$$\frac{\partial}{\partial t}\mathcal{P} + \vec{\nabla}\cdot J = 0. \tag{1.9.6}$$

Integrating this over all $\vec{r}$, $\int\left(\frac{\partial}{\partial t}\mathcal{P} + \vec{\nabla}\cdot J\right)d^3\vec{r} = \frac{d}{dt}\int\mathcal{P}d^3\vec{r} + \oint\vec{J}\cdot d\vec{a} = 0$, where we have used the divergence theorem. Observe that

$$\frac{dP}{dt} = -\oint\vec{J}\cdot d\vec{a}, \tag{1.9.7}$$

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where $P = \int \mathcal{P} d^3\vec{r}$ is the total probability in a given volume enclosed by the area integral. Thus, the total probability in a volume changes only by a flow of probability current through the enclosing area.

Question 1.9.1: Given a normalized wave function at time $t = 0$, does it stay normalized as it evolves in time according to the Schrodinger equation? (A). Yes, (B). No, (C). Depends on wave function.

Answer: The answer is (A). Yes. This is because of conservation of probability. See below for more detail.

Because the probability density for finding the particle at infinity must be zero (otherwise the wavefunction would not be normalizable), $\vec{J} = 0$ at infinity. Therefore,

$$\frac{dP}{dt} = 0. \tag{1.9.8}$$

In other words, the total probability is constant in time. If the wavefunction is normalized with $P = 1$ at one time, it stays normalized at all times.

Observe that the conservation of probability is analogous to the conservation of charge in electromagnetism, where the total charge within a given volume could only change if a net electric current flows out or in (or net charge exits or enters) through the boundary enclosing the volume.

Question 1.9.2: Which of the following are true at $x = 0$ and $t = 0$ about a system whose wavefunction changes from $\psi(x, 0)$ at time $t = 0$ to $\psi(x, \delta t)$ at time $t = \delta t$ as shown in the figure below?

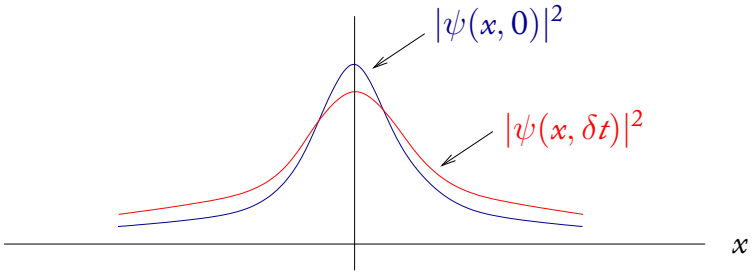


Figure 1.9.1: Wavefunction at time $t = 0$ and $t = \delta t$.

(A). $\frac{\partial \mathcal{P}(x,t)}{\partial t} > 0$, (B). $\frac{\partial \mathcal{P}(x,t)}{\partial t} < 0$, (C). $\frac{\partial J(x,t)}{\partial x} > 0$, (D). $\frac{\partial J(x,t)}{\partial x} < 0$.

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Answer: The answers are (B). $\frac{\partial \mathcal{P}(x,t)}{\partial t} < 0$ and (C). $\frac{\partial J(x,t)}{\partial x} > 0$. Note that the probability density decreases at $x = 0$ and $t = 0$ while the probability current away from the origin increases. Mathematically, $\frac{\partial}{\partial t}\mathcal{P} = -\vec{\nabla} \cdot \vec{J}$, and $\frac{\partial}{\partial t}\mathcal{P} < 0$ implies $\vec{\nabla} \cdot \vec{J} = \frac{\partial}{\partial x} J > 0$.

Exercise 1.9.1: Considering the wavefunction $\psi(x) = A \sin(kx)$, it can be rewritten as $\psi(x) = \frac{A}{2i}\left(e^{ikx} - e^{-ikx}\right) = \frac{1}{\sqrt{2}}\left(\psi_R(x) + \psi_L(x)\right)$, where $\psi_R(x) = -i\frac{A}{\sqrt{2}}e^{ikx}$ and $\psi_L(x) = i\frac{A}{\sqrt{2}}e^{-ikx}$. Show that $J(x) = 0$, $J_R(x) = v\mathcal{P}_R(x)$, and $J_L(x) = -v\mathcal{P}_L(x)$, where $v = \frac{\hbar k}{m} = \frac{p}{m}$ is velocity, for each of $\psi(x)$, $\psi_R(x)$, and $\psi_L(x)$, and point out why these currents are zero, positive, or negative.

1.10 The Schrödinger equation as an operator equation

Question 1.10.1: Does the plane wave $\psi(\vec{r}, t) = Ae^{i(\vec{k}\cdot\vec{r} - \omega t)}$, where A, $\vec{k}$, and ω are constants, solve the Schrödinger equation of a free particle with zero potential energy? (A). Yes, for $\hbar\omega = \frac{\hbar^2 k^2}{2m}$, (B). Yes, for $\hbar\omega = pc$, (C). Yes, for any relation between ω and k, (D). Not at all.

Answer: The answer is (A). Yes, for $\hbar\omega = \frac{\hbar^2 k^2}{2m}$.

With $V = 0$, the Schrödinger equation becomes $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}, t)$. The left hand side is $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = i\hbar\frac{\partial}{\partial t}\left(Ae^{i(\vec{k}\cdot\vec{r} - \omega t)}\right) = i\hbar(-i\omega)\left(Ae^{i(\vec{k}\cdot\vec{r} - \omega t)}\right) = \hbar\omega\psi(\vec{r}, t)$. The right hand side is $-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}, t) = \frac{\hbar^2\vec{k}^2}{2m}$. Therefore, the solution is solved and the answer is (A). Yes, for $\hbar\omega = \frac{\hbar^2 k^2}{2m}$. Recalling that k is the wave vector $k = \frac{2\pi}{\lambda}$ and using de Broglie's formula $\lambda = \frac{h}{p}$, we have $k = \frac{2\pi p}{h} = \frac{p}{\hbar}$ or $\vec{p} = \hbar\vec{k}$. With this, $\hbar\omega = \frac{\vec{p}^2}{2m}$, which is just the energy of the particle. Observe that $i\hbar\frac{\partial}{\partial t}\psi = E\psi$ and $-i\hbar\vec{\nabla}\psi(\vec{r}, t) = \vec{p}\psi(\vec{r}, t)$. Therefore, we can identify $E \to i\hbar\frac{\partial}{\partial t}$ and $\vec{p} \to -i\hbar\vec{\nabla}$.

For contrast, ψ is a solution to the wave equation, $\frac{\partial^2}{\partial t^2}\psi(\vec{r}, t) = c^2\vec{\nabla}^2\psi(\vec{r}, t)$, with $\hbar\omega = pc = E$.

In other words, the energy-momentum relation of a Newtonian massive particle is in the Schrödinger equation while that energy-momentum relation of a photon is in the wave equation but not the Schrödinger equation.

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Rewriting the Schrödinger equation as

$$\left(i\hbar\frac{\partial}{\partial t}\right)\psi(\vec{r},t) = \left(-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{r},t)\right)\psi(\vec{r},t), \tag{1.10.1}$$

we see that the Schrödinger equation can be viewed as an eigenvalue problem in linear algebra,

$$E\psi(\vec{r},t) = \hat{H}\psi(\vec{r},t), \tag{1.10.2}$$

where

$$E \to i\hbar\frac{\partial}{\partial t} \tag{1.10.3}$$

and

$$\hat{H} = \frac{\hat{\vec{p}}^2}{2m} + \hat{V}(\hat{x},t) \tag{1.10.4}$$

with

$$\hat{p} \to -i\hbar\vec{\nabla}, \tag{1.10.5}$$

$$\hat{\vec{r}} \to \vec{r}, \tag{1.10.6}$$

and

$$\hat{V} \to V(\vec{r},t). \tag{1.10.7}$$

The hat is put to remind us that the corresponding objects are operators, rather than numbers. With these, and keeping the time derivative on the left hand-side, we can rewrite the Schödinger equation as

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi. \tag{1.10.8}$$

Observe that $H = \frac{\vec{p}^2}{2m} + V$ is the total energy of the particle in classical mechanics. H is called the Hamiltonian and $\hat{H}$ is called the Hamiltonian operator. Moreover, $\hat{\vec{r}}$ is called the position operator, $\hat{\vec{p}}$ is called the momentum operator and the substitution $\hat{\vec{p}} \to -i\hbar\vec{\nabla}$ is called the Dirac substitution. Observe that $V(\vec{r},t)$ is potential energy and it is real.

In general, the procedure for constructing the Hamiltonian operator for a given system is as follows. Write down the classical Hamiltonian $H(\vec{p},\vec{r},t)$. Incorporate purely quantum mechanical effects (such as spin). Promote the classical Hamiltonian to an operator with $\vec{p} \to \hat{\vec{p}}$ and $\vec{r} \to \hat{\vec{r}}$. Time is not an operator but a label in quantum mechanics. With $\hat{H}$ in hand, we write the Schrödinger equation, $i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$.

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